

UEI515: ANALOG AND DIGITAL COMMUNICATION

L	T	P	Cr
3	0	0	3.0

Course objective: The main objectives of this course are to acquire knowledge about analog communication systems.

Unit I Introduction: Introduction to communication systems: Modulation, type and need for modulation. Introduction to Analog communication, Introduction to Digital communication

Unit II Amplitude and SSB modulation: Theory of amplitude modulation, AM transmission, , Basic principle of AM generation, Square law modulation, AM reception, Super heterodyne Receiver, Introduction and advantages of SSB Transmission, Generation of SSB, The Phase Shift Method, Vestigial Side-band Modulation, application of amplitude modulation.

Unit III Angle modulation: Theory of frequency modulation and demodulation Narrow band FM, Wide band FM, Phase modulation, Phase modulation obtained from frequency modulation, comparison of various analog communication system (AM-FM-PM)

Unit IV Analog Pulse Modulation: Introduction, Pulse amplitude modulation (PAM), Pulse Time Modulation (PTM), Pulse Width Modulation (PWM), Pulse Position Modulation (PPM).

UNIT V Digital Pulse Modulation Systems: Pulse code modulation (PCM), Noise in PCM systems, Differential pulse code modulation (DPCM), Adaptive pulse code modulation (ADPCM), Delta modulation (DM), Comparison of PCM, DPCM and DM, Adaptive delta modulation.

UNIT VI Digital Modulation Techniques: Amplitude shift keying, frequency shift keying, phase shift keying. Quadrature amplitude modulation, Bandwidth efficiency, comparison of various communication techniques (ASK, FSK, PSK, QAM)

After the completion of the course student will be able to:

1. Demonstrate the knowledge of Amplitude and SSB modulation
2. Demonstrate the knowledge of angle modulation
3. Elucidate the various analog and digital pulse modulation schemes
4. Elucidate digital modulation techniques and their comparison in communication system

Text Books: Lathi.B.P., *Modern Digital and Analog Communications Systems* 3rd ed.

Reference Books:

1. Taub, H., *Principles of Communication Systems*, McGraw-Hill (2008) 3rd ed.
2. Haykin, S., *Communication Systems*, John Wiley (2009) 4th ed.
3. Proakis, J. G. and Salehi, M., *Fundamentals of Communication Systems*, Dorling Kindersley (2008) 2nd ed.
4. Kennedy, G., *Electronic Communication Systems*, McGraw-Hill (2008) 4th ed.
5. Proakis John G., *Digital Communication System*, McGraw, (2000) 4th ed.
6. Simon Haylein, *Digital Communication Systems*, Wiley India edition, (2009) 2nd ed.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30